

Validating Scintillation in CHIME/FRB Data

Conditions, evidence, and open issues for the method ratification (wayfinder ticket 02)

Ground truth: analysis/window-tuning-campaign-2026-07-17/ at FLITS main 666306d1 · staged for Faber2026 by PR #140

Why CHIME-band scintillation is hard

Upchannelized baseband

Δv_d sits near/below the native resolution; measurements ride the upchannelization chain, not raw channels.

Coarse-channel bandpass

Multiplicative common-mode structure within each 390.625 kHz coarse channel — shown to live upstream in the baseband (windowing left its amplitude unchanged, 0.586 $\rightarrow$ 0.62–0.68).

Data lineage

A gen-1 de-chirp defect contaminated early upchannelized products; every input must be gen-2+, checksum-provenanced.

Scattering-tail chromatics

The tail carries broad spectral structure distinct from the narrow core scintle — window choice alone can inflate γ by 2.6–3.6 $\times$.

A frequency ACF cannot tell these apart unaided

Every failed route below died on one of these confounds — the validation conditions exist to force the separation.

The route graveyard — and what each failure bought

Route	Predeclared gate	Outcome	Lesson
Cross-fitted all-pairs estimator (C1)	Low-modulation calibration cells + family-wise null	0/8 cells passed; nulls failed	Estimator-level cleverness cannot beat the common mode
Windowed fine-channelization (P1)	10× common-mode suppression, 5 predeclared variants	0/5; amplitude unchanged → structure is upstream in baseband	Located the confound; killed re-channelization fixes
Ratio statistic, Route B (P2)	Cancellation (G2) + sensitivity (G1)	G2 PASS ~100×; G1 FAIL	Cancelling the mode also cancels the signal
Delay-domain optimal estimator (P3)	$\geq 3\sigma$ forecast; frozen gates; one-shot unblinding	Calibrated; unblinded structure declined as intrinsic envelope	A calibrated estimator exists; attribution is the bottleneck
Envelope modeling (P4)	Predeclared grids + surrogate certification	DOCUMENTED-FAIL: 13/14 operating points fail mismatch control	Intrinsic envelope not separable post hoc — must be handled in-window

Design requirements the failures forced

Predeclare, then look

Gates, thresholds, and status rules frozen before burst data is inspected — every route above did this; the survivor must too.

Null-first

An off-pulse null distribution accompanies every measurement; a detection must beat its own null, not a theory curve.

Carry the window systematic

Estimator variants are reported, not hidden — the window-choice spread ($\gamma_{\text{win_sys}}$) rides every quoted γ .

Decide by injection, not assertion

Estimator and model-selection thresholds chosen by known-truth injections with measured false-positive rates.

Mechanical status vocabulary

detection / diagnostic_only / non_detection assigned by rule, with the basis string stored next to the number.

Two-band consistency as the external check

CHIME $\gamma(v)$ scaling must cohere with the DSA band — the cross-check no single-band route could offer.

The candidate method — windowed refit, two-band scaling

1 Coherently dedispersed, de-scalloped dynamic spectrum (gen-2+, checksummed)

2 Window selection: matched-significance burst core + pre-burst off-window

3 Per-subband frequency ACF (4 subbands)

4 2-Lorentzian vs 1-Lorentzian model selection ($\Delta\text{BIC} \geq 6$)

5 $\gamma(\nu)$ per resolved subband, with scintle + window error terms

6 Scaling index α across resolved subbands; mechanical status rule

Key choices are injection-decided

- Core-boxcar window: on clean truth all three estimators agree to ~2%; with tail chromatics, tail-expanded inflates γ 2.6–3.6 $\times$, matched-weight 1.3–1.9 $\times$ — core stays unbiased.
- Pre-burst off-window: post-burst windows sit in the scattering tail and corrupt per-channel gains. For chromatics this is the difference between a 4-subband detection and a railed 2-subband collapse.
- Grid regularization ON — fitted grids uniform, absolute γ quotable.

Calibration: the fitter earns its thresholds

9/9

injection-recovery grid PASS — γ and inner components inside 90% CIs on known truth

0.02–3 MHz

range over which the 2-component model recovers γ unbiased; the single Lorentzian fails a narrow scintle under a broad envelope

0/150 + 0/150

false-positive rate of the $\Delta\text{BIC} \geq 6$ two-component gate on envelope-only and pure-noise truth sets

$\gamma_{\text{win_sys}}$, γ_{tail}

retained estimator variants keep the window systematic visible in every record, per subband

The predeclared validation conditions (verbatim from the records)

detection

$n_{\text{resolved_physical}} \geq 3$ AND $\alpha \in (1.5, 6.0)$ open interval AND off-pulse null not majority-fail

diagnostic_only

subbands resolve but the slope is unphysical — e.g. hamilton: $n=2$ resolved, $\alpha = +8.94$ outside $(1.5, 6.0)$

non_detection

$n_{\text{resolved_physical}} < 2$ — no measurable slope; no α is quoted at all

Per-subband 'resolved' is itself gated: amplitude S/N, shape check, and the ΔBIC line test must all pass; every γ carries fit, scintle, and window-systematic error terms. The basis string is stored beside every verdict — the rule is auditable per burst.

Open question for ratification: the $(1.5, 6.0)$ window — thin-screen Kolmogorov expects $\alpha \approx 4.4$; the permissive lower bound admits shallow scalings. Predeclared, but is it the interval you want to defend to a referee?

The 12-burst campaign outcome



chromatica

$\alpha = +1.72 \pm 0.40$ (n = 4)



zach

$\alpha = +3.03 \pm 0.65$ (n = 3)



hamilton

$\alpha = +8.94 \pm 3.22$ — unphysical slope, correctly quarantined



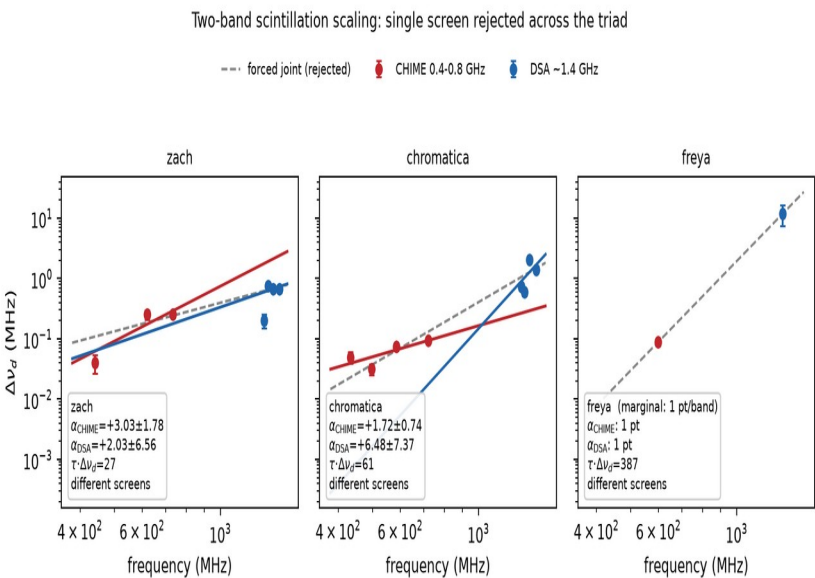
casey, freya, phineas, wilhelm

1 resolved subband each — below the 2-subband floor



isha, johndoell, mahi, oran, whitney

0 resolved subbands

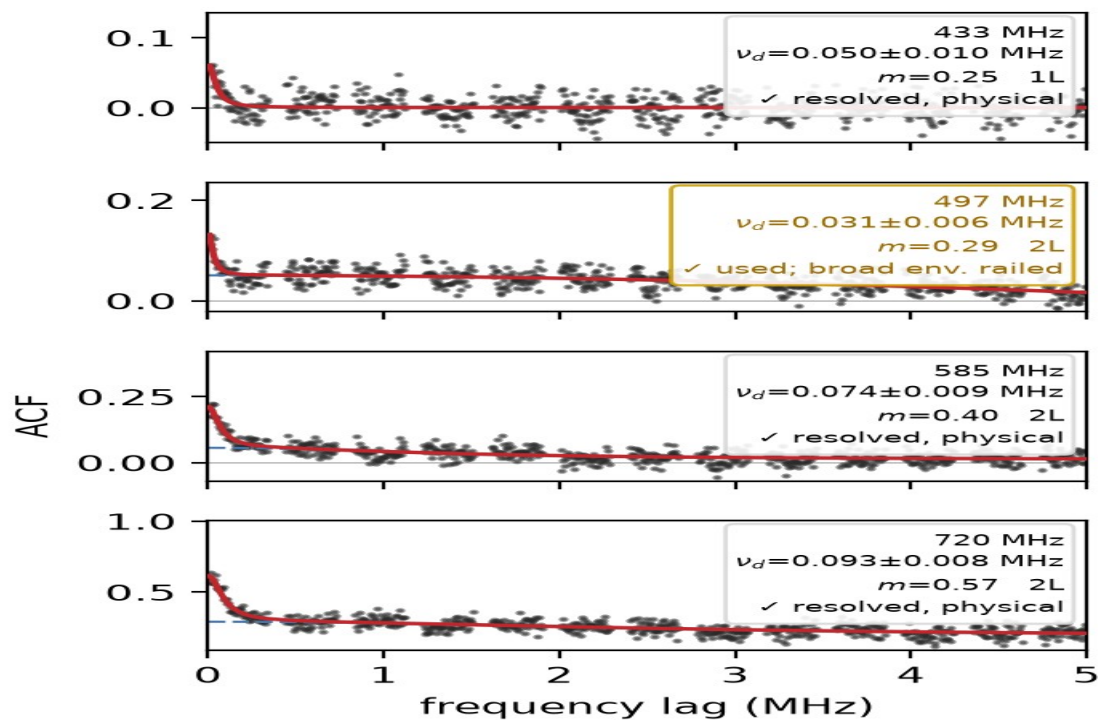


Two-band scaling summary (staged for the manuscript in PR #140)

Full CHIME sample coverage: this campaign already ran all twelve co-detections — ratification does not ‘unblock’ a future CHIME run; it decides whether THIS one is citable.

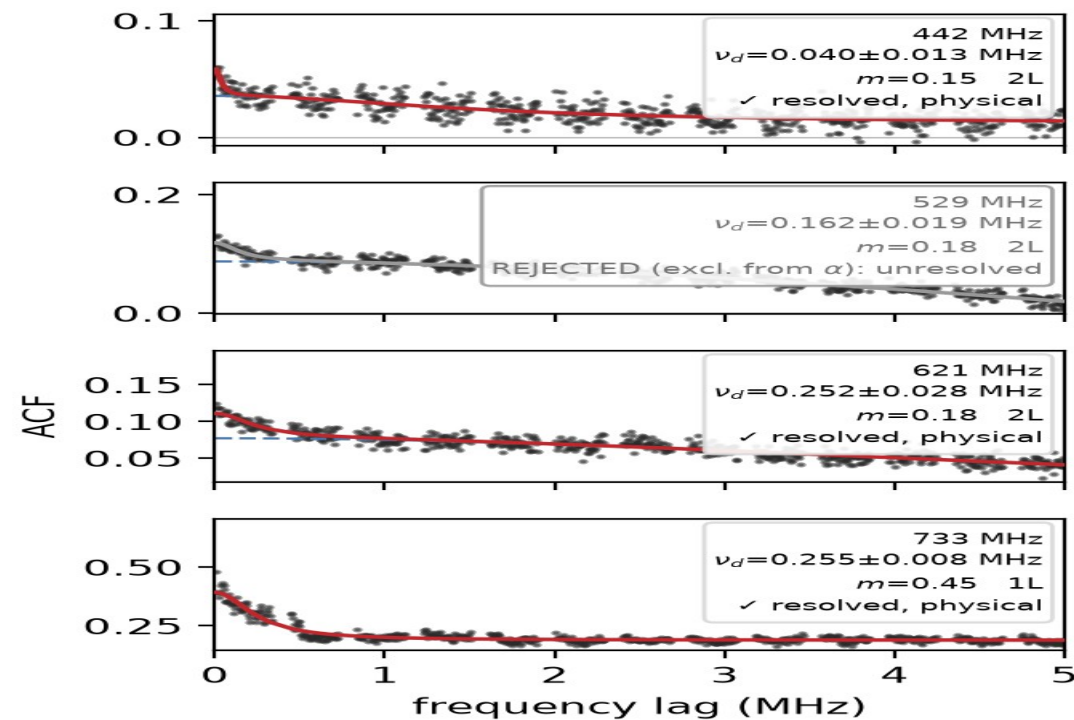
The two detections

chromatica: CHIME scintillation ACF ladder
 $\alpha = +1.72 \pm 0.40$ (n=4)



chromatica — 4 resolved subbands, pre-burst window; $\alpha = +1.72 \pm 0.40$

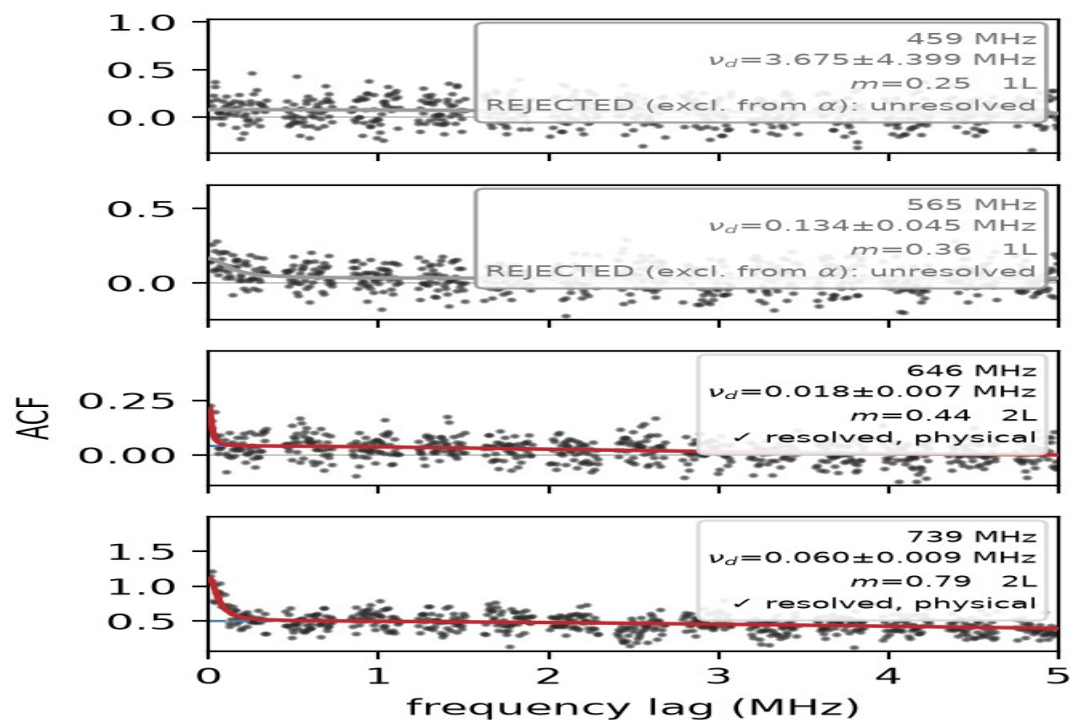
zach: CHIME scintillation ACF ladder
 $\alpha = +3.03 \pm 0.65$ (n=3)



zach — 3 resolved subbands; $\alpha = +3.03 \pm 0.65$

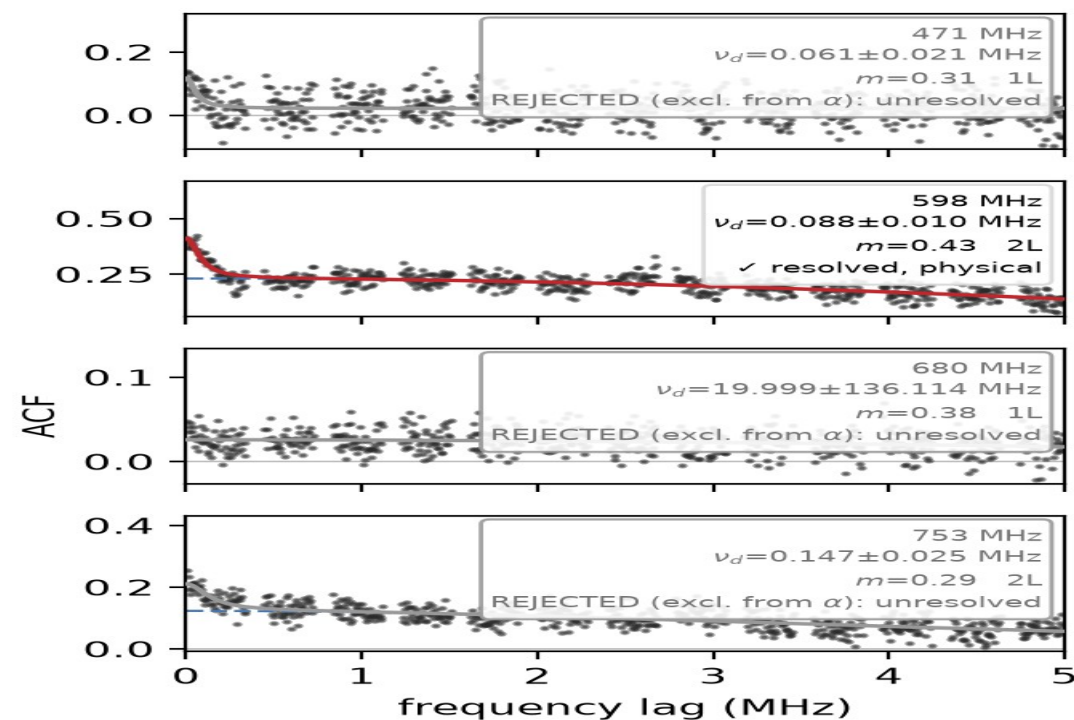
The rules rejecting what should be rejected

hamilton: CHIME scintillation ACF ladder
 $\alpha = +8.94 \pm 3.22$ (n=2)



hamilton — 2 subbands resolve but $\alpha = +8.94$ is unphysical $\rightarrow$ diagnostic_only, no science claim

freya: CHIME scintillation ACF ladder
 α : n/a (insufficient resolved physical subbands)



freya — the burst that failed four qualification routes lands as non_detection here (1 resolved subband): the new method agrees with the graveyard rather than contradicting it

What this ratification must NOT cover: $\tau \cdot \Delta v_d$

The two-screen products ($\tau \cdot \Delta v_d = 26.7 / 61.0 / 386.7 / 1.3$) are composite: Δv_d from this campaign, τ from the scattering mass-refit — which is still pre-ratification (leakage-injection verdict, count adoption, and validation all pending).

- Ratifying $\tau \cdot \Delta v_d$ now would smuggle unratified τ values into citability through a product.
- Recommendation: Δv_d side citable with this ratification; $\tau \cdot \Delta v_d$ and every screen-placement inference stay pending until the scattering side ratifies (free- α reporting + count remediation tickets).
- The scint-to-scattering coupling design (constraint layer, prior-odds geometry) consumes these as frozen posterior/limit products once both sides are trusted — sequencing is already on the map.

Open issue found in this review: the closure record is stale

validation.json (campaign closure)

#192-era: status=closed, 24 records (with _hi variants), n_measurements = 1 (chromatica_hi), figure_gate = FAIL, long artifact-failure list

campaign_results.jsonl (beside it)

R5/R6: 12 records, TWO detections, different status vocabulary (detection vs measurement)

- The campaign's own finalization machinery (finalize_campaign.py: artifact validation + figure-review gates → promotion) has not been rerun over the R5/R6 records.
- The five staged manuscript figures (PR #140) have not passed the hash-bound review the figure gate encodes.
- The manuscript-side loader/tests still encode the #192 contract — the pin bump fails CI until the sync (diagnosed on PR #140).
- The staged two-band figure quotes larger alpha errors than the records (zach ± 1.78 vs ± 0.65 ; chromatica ± 0.74 vs ± 0.40) — presumably a systematics-inflated convention; the convention must be named and made consistent before either number is quoted.
- **Ratification should therefore carry CONDITIONS PRECEDENT: rerun closure + figure review over R5/R6; regenerate validation.json; reconcile the error conventions; then the manuscript contract sync.**

Mapping to the five-term re-trust contract (per decision (a))

(i) Verified input lineage	MET (conditional)	gen-2+ inputs, checksummed; confirm per-burst lineage table at closure rerun
(ii) Injection recovery of known truth	MET	recover + discriminate harnesses; 9/9 grid; estimator chosen by injection
(iii) Rail = rejection, never a limit	MET by construction	γ rails $\rightarrow$ subband unresolved $\rightarrow$ excluded from α ; no railed value is quoted
(iv) Posterior-predictive / adequacy check	PARTIAL	Δ BIC + shape gates play this role; no explicit PPC on the ACF fits — note for wf-13
(v) Independent cross-check, re-derived	PARTIAL	two-band consistency is the check; DSA-band re-runs under the contract still owed

Two PARTIALs are the honest gaps wf-13 inherits — they are recorded, not hidden, and neither invalidates the predeclared-gate basis chosen for this ratification.

The decision on the table

RATIFY — the method

Windowed refit + core-boxcar + pre-burst off-window + Δ BIC model selection + mechanical status rules, as the qualifying CHIME-band route.

RATIFY — the verdicts

The 12-burst status set and the two α scalings (chromatica $+1.72 \pm 0.40$, zach $+3.03 \pm 0.65$), citable once the conditions precedent clear.

EXCLUDE — $\tau \cdot \Delta v_d$

Composite with unratified τ ; stays pending behind the scattering-side ratification.

CONDITIONS PRECEDENT

Rerun campaign closure + hash-bound figure review over R5/R6; regenerate validation.json; manuscript contract sync (turns PR #140 green). Decide the α -interval defense (1.5, 6.0).

Everything above is grounded in the pinned artifacts at FLITS 666306d1; nothing depends on this session's summary being right.